CUSTIQ FORCE

E-commerce Customer Intelligence & Retention

DEPI Graduation Project — Data Science

Project information	Details
Project name	Custiq Force
Supervisor	Eng. Ahmed Mostafa
Project Lead	Mohanned Mohammed Galal
Team members	Shahd Magdy Abdelwahd; Marwa Elbadawy Mohammed; Yassin Hassan Abdelkader; Mohammed Atef Mosa
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Document version	Draft v1.1 — Parts 1–3
Dataset	Olist Brazilian E-Commerce Public Dataset

This document covers Project Planning & Management, Literature Review, and Requirements Gathering. It describes planned work only; project findings and model performance will be added after analysis.

1. Project Planning & Management

1.1 Project Overview

Custiq Force is a Data Science graduation project focused on turning e-commerce transaction and customer data into useful business insights. The project combines data validation, SQL-based business metrics, Python analysis, machine learning, and an interactive Streamlit dashboard. Its main areas are repeat-purchase behavior, customer value, and identifying customers who may be at risk of not returning.

1.2 Problem Statement

E-commerce data is often spread across related tables. Incorrect customer identification or joins can distort order counts and revenue measures. Businesses may also find it difficult to identify meaningful patterns in customer activity and decide where retention analysis may be useful. Custiq Force will build a documented workflow to validate data, calculate transparent metrics, evaluate predictive models where feasible, and present findings responsibly.

1.3 Aim and Objectives

Aim: Build an end-to-end customer intelligence prototype for e-commerce retention analysis using the Olist Brazilian E-Commerce Public Dataset.

- Inspect the downloaded files and document their schema, record counts, date coverage, quality, and relationships.
- Clean and prepare data through reproducible Python scripts or notebooks.
- Create documented SQL queries and business metrics for customer and order analysis.
- Define a defensible repeat-purchase or at-risk prediction target if the data supports it.
- Train a baseline machine-learning model, compare suitable alternatives, and evaluate results on held-out data.
- Interpret model outputs and explain their limitations in business terms.
- Build a Streamlit dashboard presenting selected validated metrics, analyses, model outputs, and recommendations.
- Maintain one GitHub repository containing code, documentation, and setup instructions.

1.4 Dataset

The selected source is the Olist Brazilian E-Commerce Public Dataset on Kaggle. Its published description reports approximately 100,000 orders from 2016 to 2018, with related customer, order, item, payment, review, product, seller, and geolocation information. The team must verify the exact files and figures in its own downloaded copy before reporting them as project findings. [1]

For repeat-purchase analysis, the team should investigate customer_unique_id, which is intended to link activity for the same underlying customer. customer_id identifies the customer record associated with an order and should not automatically be treated as a persistent person-level identifier. [2]

1.5 Project Scope

Included in the project	Out of scope unless approved
Dataset inventory, data dictionary, and quality checks	Live production e-commerce integration
Data cleaning, EDA, and visualizations	Automated marketing campaigns or direct customer messaging
SQL queries and documented business metrics	Claims of causal impact without appropriate experiments
ML target definition, baseline, comparison, and evaluation	Predictions or features unsupported by available data
Streamlit dashboard and evidence-based recommendations	Production-grade customer management system
GitHub repository, setup guide, tests, and documentation	Raw-data redistribution without license review

1.6 Technology Stack

Technology	Purpose
Python and Pandas	Data preparation, analysis, and feature engineering
SQL / MySQL	Data storage, joins, validation, and business metrics where suitable
Scikit-learn	Model pipelines, training, and evaluation
Matplotlib / Seaborn	Exploratory and explanatory visualizations
Streamlit	Interactive dashboard
GitHub	Version control, collaboration, and documentation

1.7 Timeline and Milestones

The documentation deadline for the first three sections is 19 October 2026. The planned final project submission date is 6 December 2026. No intermediate dates were supplied, so milestones are listed in execution order without inventing deadlines.

Milestone	Main activities	Completion evidence
1. Data validation	Inventory files; inspect schema, dates, keys, counts, and license	Dataset inventory, data dictionary, decision log
2. Database and preparation	Design schema; load suitable tables; clean data; test joins	Reproducible scripts and quality checks
3. EDA and metrics	Analyze customer activity, order status, and supported value measures	EDA outputs, SQL queries, metric definitions
4. Target and features	Choose cutoff and windows; prevent future-data leakage	Target specification and feature list
5. Modeling	Train baseline and candidate models; inspect errors	Evaluation report and interpretation
6. Dashboard	Build Streamlit pages from validated outputs	Working dashboard and run instructions
7. Final review	Test calculations, usability, documentation, and presentation	Reviewed repository and final presentation

1.8 Team Roles and Responsibilities

Each member has a primary workstream, while the full team remains responsible for reviewing assumptions, code, and findings.

Team member	Primary responsibility
Mohanned Mohammed Galal	Project Lead; SQL, business metrics, and GitHub coordination
Shahd Magdy Abdelwahd	Dataset preparation and database setup
Marwa Elbadawy Mohammed	Data cleaning, EDA, and customer behavior analysis
Yassin Hassan Abdelkader	Machine learning, model evaluation, and interpretation
Mohammed Atef Mosa	Streamlit dashboard, results, and recommendations

1.9 Risk Management

Risk	Mitigation
Customer identity confusion	Validate identifiers; use customer_unique_id for person-level repeat analysis where appropriate.
Data leakage	Set a prediction cutoff and exclude features derived from later information.
Join duplication	Document each table's record grain and test join cardinality and row counts.
Unclear at-risk definition	Choose an inactivity window after reviewing purchase cadence and date coverage.
Class imbalance	Inspect target distribution and use metrics beyond accuracy when appropriate.
Poor data quality	Document missing values, exclusions, assumptions, and checks.
Overstated recommendations	Distinguish prediction and correlation from causal impact.

Scope creep	Prioritize a complete baseline before optional advanced work.
Dataset licensing	Review license terms before sharing raw data.
Team integration issues	Use one repository, issues, descriptive commits, and code review.

1.10 Success Criteria

Area	Success indicator	Verification
Data understanding	Files and key tables documented	Review inventory and data dictionary
Data quality	Critical quality checks recorded	Review outputs and row-count comparisons
Business metrics	Formulas and order-status rules documented	Cross-check SQL and Python calculations
Reproducibility	A teammate can rerun core steps	Test setup instructions
ML validity	Defensible target and held-out evaluation	Review target, split, and metrics
Dashboard	Clear labels and usable navigation	Team walkthrough and feedback
Delivery	Repository, documentation, dashboard, and presentation ready	Final checklist

1.11 GitHub Repository Structure

```
custiq-force/\n├── README.md\n├── docs/\n│   ├── project_documentation.pdf\n│   └── data_dictionary.md\n├── data/\n│   └── README.md\n├── notebooks/\n├── src/\n├── dashboard/\n├── tests/\n└── requirements.txt
```

Do not commit credentials, tokens, or private information. Check the dataset license before uploading raw files. The README should explain the project, dataset source, environment setup, and how to run the analysis and dashboard.

2. Literature Review

2.1 Purpose

This review considers related approaches to customer value analysis, customer segmentation, churn prediction, and dashboard-based decision support. These sources inform the project design; results reported by other researchers are not results for Custiq Force.

2.2 Related Datasets and Methods

Olist Brazilian E-Commerce Public Dataset. The dataset connects orders with customers, items, payments, reviews, products, sellers, and geolocation information. This breadth supports several analyses, but the multi-table structure creates a risk of duplicated records when joins are not designed carefully. The published description reports around 100,000 orders from 2016–2018; the team must verify its local copy. [1]

RFM analysis and Online Retail. The UCI Online Retail dataset is a transactional dataset used for retail analysis. RFM summarizes Recency (how recently a customer purchased), Frequency (how often they purchased), and Monetary value (how much they spent). Chen, Sain, and Guo (2012) discuss RFM-based customer segmentation for online retail. RFM is interpretable and useful for descriptive customer-value analysis, but it does not by itself predict future purchases. [3]

Machine-learning churn prediction. Research has explored supervised models for identifying customers who may stop purchasing. A 2026 study examines an enhanced adaptive boosting approach and discusses class-imbalance challenges. This supports evaluating precision, recall, F1-score, and other suitable metrics rather than relying only on accuracy. Results from that study cannot be assumed for Custiq Force because the data and target definitions may differ. [4]

Hybrid RFM, clustering, and deep learning. A 2025 study combines RFM features, K-means segmentation, and neural-network methods for customer retention. It suggests that customer segments may complement prediction,

but complex methods can be harder to explain and validate. Custiq Force will prioritize a transparent baseline before optional advanced methods. [5]

Integrated analytics and dashboard decision support. A 2026 open-access article discusses combining SQL-based integration, customer segmentation, churn prediction, and dashboard reporting using e-commerce data. This supports connecting data preparation, model evaluation, and reporting while clearly defining Custiq Force's own target and methodology. [6]

2.3 Comparison of Related Approaches

Approach	Strengths	Limitations	Lesson for Custiq Force
Olist dataset [1]	Multiple business entities and transaction details	Complex joins; local counts need verification	Document table grain and test joins
RFM / Online Retail [3]	Simple, interpretable customer summaries	Descriptive; results depend on business rules	Use transparent RFM-like features where supported
Churn prediction [4]	Can estimate risk for prioritization	Sensitive to labels, imbalance, and leakage	Use a time-aware target and multiple metrics
RFM + clustering + deep learning [5]	Combines customer segments with prediction	Higher complexity and explanation burden	Treat advanced methods as optional extensions
Integrated dashboard [6]	Connects metrics, modeling, and decision support	Study results may not generalize	Use Custiq Force's own validated outputs

2.4 Research Gap and Project Positioning

Related work shows that RFM analysis, segmentation, churn prediction, and dashboards can each support retention decisions. Custiq Force will focus on integrating these elements into a reproducible student project using the Olist dataset. Its contribution is not a claim of a new algorithm; it is the careful connection of data validation, documented business metrics, time-aware modeling, transparent evaluation, and a practical dashboard.

The project will investigate the following questions if the dataset supports them:

- What patterns are visible in customer purchase activity and order history?
- How can customer activity and value be summarized using transparent measures?
- Can the available timestamps support a defensible repeat-purchase or at-risk target?
- Which model offers a useful balance between identifying at-risk customers and generating false alerts?
- How can the dashboard communicate findings and uncertainty clearly?

2.5 Literature Review Conclusion

The literature supports starting with data quality and interpretable metrics before applying machine learning. Custiq Force will follow that sequence and will not present results from other studies as its own.

3. Requirements Gathering

3.1 Stakeholders and Needs

Stakeholder	Needs	Project response
Business decision-makers	Understand purchase patterns, customer value, and possible retention priorities	Clear metrics, risk indicators, and evidence-based recommendations
Retention / marketing analysts	Review customer groups for possible engagement	Interpretable summaries and risk indicators; no automated messaging
Data analysts / data scientists	Reproduce calculations, inspect data, and evaluate models	Data dictionary, scripts, SQL queries, feature definitions, and evaluation
Supervisor and DEPI evaluators	Assess methodology, technical quality, teamwork, and documentation	Traceable decisions, repository, dashboard, and final presentation
Project team	Coordinate work and integrate components	Single repository, clear roles, issues, and setup instructions

3.2 Functional Requirements

The following priorities are proposed for planning and may be refined with the supervisor.

ID	Requirement	Priority
FR-01	Inventory downloaded files and report actual row counts, columns, data types, and date coverage.	Must
FR-02	Check missing values, duplicates, timestamps, unusual values, key uniqueness, and table relationships.	Must
FR-03	Document table grain and validate joins to prevent unintended row multiplication.	Must
FR-04	Prepare analysis-ready data using reproducible Python scripts or notebooks.	Must
FR-05	Calculate agreed customer and order metrics with documented formulas and order-status rules.	Must
FR-06	Perform EDA and produce visualizations for defined business questions.	Must
FR-07	Define the ML prediction unit, cutoff, windows, and target where supported by data.	Must for ML
FR-08	Train a baseline model and compare suitable alternatives using held-out data.	Must for ML
FR-09	Report appropriate evaluation metrics and interpret errors and limitations.	Must for ML
FR-10	Build a Streamlit dashboard for selected validated outputs.	Must
FR-11	Present recommendations while distinguishing observed patterns from predictions and hypotheses.	Must
FR-12	Maintain source code, documentation, and setup instructions in one GitHub repository.	Must
FR-13	Add optional filters or advanced methods if time and data quality permit.	Could

3.3 Non-Functional Requirements

ID	Quality attribute	Requirement
NFR-01	Correctness	Metrics follow documented formulas; joins and key totals are checked.
NFR-02	Reproducibility	Code, dependencies, assumptions, and execution steps are documented.
NFR-03	Performance	Analysis and dashboard run on the team's available hardware using validated data.
NFR-04	Usability	Charts, filters, labels, units, and risk indicators are clear to non-technical reviewers.
NFR-05	Explainability	The team explains model choice, prediction meaning, and limitations.
NFR-06	Security and privacy	Never commit credentials or private information; respect dataset license terms.
NFR-07	Maintainability	Organize preparation, metrics, modeling, and dashboard code into understandable modules.
NFR-08	Robustness	Handle missing or unexpected values explicitly and avoid silently displaying invalid results.
NFR-09	Collaboration	Use one repository, descriptive commits, issue tracking, and code review.
NFR-10	Responsible reporting	Do not treat a risk score as certainty or correlation as proof of causation.

3.4 Data Requirements and Business Rules

- Source verification: confirm the actual files, columns, row counts, and date coverage.
- Customer identity: validate customer_id and customer_unique_id relationships before customer-level analysis.
- Qualifying orders: agree which order statuses count for each metric and how canceled or unavailable orders are handled.
- Financial measures: specify whether order value includes item prices, freight, or other supported amounts; avoid double-counting items or payments.
- Time-aware prediction: use only information available by the prediction cutoff to build features.
- At-risk definition: choose an inactivity window based on observed date coverage and purchase cadence; document it as a project-specific definition.
- Limitations: the historical dataset may not include marketing exposure, campaign costs, customer intent, or all factors affecting repeat purchases.

3.5 Machine-Learning Requirements

- Define the business question, prediction unit, historical feature window, cutoff, and future outcome window.
- Create features using information available at the cutoff only.
- Inspect class balance and establish a simple baseline before comparing candidate models.
- Use a held-out test set and a time-based split where feasible.
- Report precision, recall, F1-score, confusion matrix, and ROC-AUC when appropriate.
- Inspect errors and explain the trade-off between missed at-risk customers and false alerts.
- Assess probability calibration if predicted scores will be used as probabilities and if feasible.
- Explain that feature importance describes model associations and does not prove causation.

3.6 Dashboard Requirements

View	Proposed content
Overview	Validated order/customer metrics, dataset coverage, and key caveats
Customer behavior	Repeat-purchase patterns, recency/frequency summaries, and value analysis where supported
Order and product analysis	Order status, product category, payment, freight, or review summaries when relevant
At-risk indicators	Agreed inactivity definition and/or model outputs, clearly labeled as estimates
Model evaluation	Model, test design, metrics, confusion matrix, and limitations
Recommendations	Evidence-linked suggestions that distinguish observed patterns from hypotheses

3.7 Assumptions and Constraints

- The team can access the Olist dataset and use it under its license conditions.
- The data may support customer-level analysis, but the repeat-purchase target must be validated.
- The team has access to Python, a SQL database if needed, Streamlit, and GitHub.
- The result is a decision-support prototype, not a production service or automated marketing system.
- The team will coordinate intermediate deadlines internally because none were provided in the project brief.

3.8 Overall Acceptance Criteria

- The GitHub repository contains clear setup instructions, source code, documentation, and the dataset source note.
- Data validation and cleaning decisions can be reproduced and are documented.

- SQL/Python metrics have explicit formulas and correct treatment of customer identity, order statuses, and joins.
- The ML component has a defensible target, leakage-aware features, held-out evaluation, and an honest interpretation.
- The Streamlit dashboard runs using documented steps and presents validated results without overstating certainty.
- The final presentation explains the business problem, data, methodology, findings, model evaluation, recommendations, and limitations.

4. Open Decisions

The team should resolve these items after inspecting the dataset and consulting the supervisor:

- Confirm Custiq Force as the final project name.
- Confirm the exact downloaded files and row counts.
- Finalize qualifying order statuses and treatment of canceled/unavailable orders.
- Choose the prediction cutoff, feature window, outcome window, and repeat-purchase definition.
- Choose and justify the inactivity window for at-risk analysis.
- Finalize customer-value and business metric formulas, including freight and payment treatment.
- Select model candidates and evaluation metrics after reviewing the target distribution.
- Confirm the GitHub repository URL and whether raw data may be shared.
- Finalize dashboard pages and presentation structure.

5. References

[1] Olist and collaborators. Brazilian E-Commerce Public Dataset by Olist. Kaggle. https://www.kaggle.com/datasets/olistbr/brazilian-ecommerce

[2] Olist and collaborators. Dataset metadata — Customers Dataset. Kaggle. https://www.kaggle.com/datasets/olistbr/brazilian-ecommerce/metadata

[3] Chen, D., Sain, S. L., & Guo, K. (2012). Data mining for the online retail industry: A case study of RFM model-based customer segmentation using data mining. Journal of Database Marketing & Customer Strategy Management, 19(3). UCI Online Retail dataset: https://archive.ics.uci.edu/dataset/352/online+retail

[4] Hang, J. (2026). Customer churn prediction model for E-commerce using an enhanced adaptive boosting algorithm. Discover Applied Sciences. https://link.springer.com/article/10.1007/s42452-026-08745-4

[5] Zaghloul, M., Barakat, S., & Rezk, A. (2025). Enhancing customer retention in Online Retail through churn prediction: A hybrid RFM, K-means, and deep neural network approach. Expert Systems with Applications. https://www.sciencedirect.com/science/article/pii/S0957417425020846

[6] Integrating Business Intelligence and CRM Systems With a Machine Learning Approach for Predictive Customer Retention in E-Commerce. (2026). PubMed Central. https://pmc.ncbi.nlm.nih.gov/articles/PMC13216662/

[7] Streamlit. Official Documentation. https://docs.streamlit.io/

References [4]–[6] inform design choices. Their findings and scores belong to those studies and are not Custiq Force results.

Conclusion

Custiq Force will connect data validation, SQL and Python analysis, machine-learning evaluation, and a Streamlit dashboard in a reproducible workflow. Its quality will depend on clear business definitions, correct table integration, leakage-aware evaluation, and transparent limitations. Final decisions and findings will be added after the team inspects the actual data.